Power

File: Power.kicad_sch

Square Wave Generator

File: Square_Wave_Generator.kicad_sch

1Hz Square Wave Generator

File: 1Hz_Square_Wave_Generator.kicad_sch

Seconds Counter

File: Seconds_Counter.kicad_sch

Minutes Counter

File: Minutes_Counter.kicad_sch

Hours Counter

File: Hours_Counter.kicad_sch

Colon LEDs

File: Colon LEDs.kicad_sch

Time Set Buttons

File: Time_Set_Buttons.kicad_sch

Hours Display

File: Hours_Display.kicad_sch

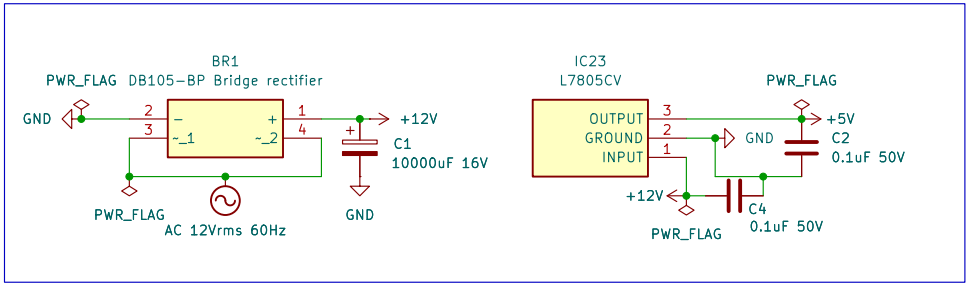
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /
File: Breadboard_Clock.kicad_sch

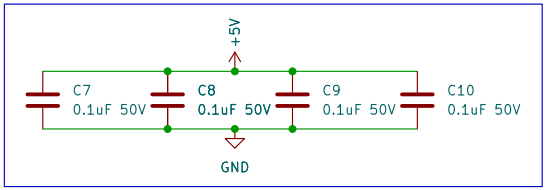
Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/10

12V DC and 5V DC



Extra Power Rail Noise Reduction



https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
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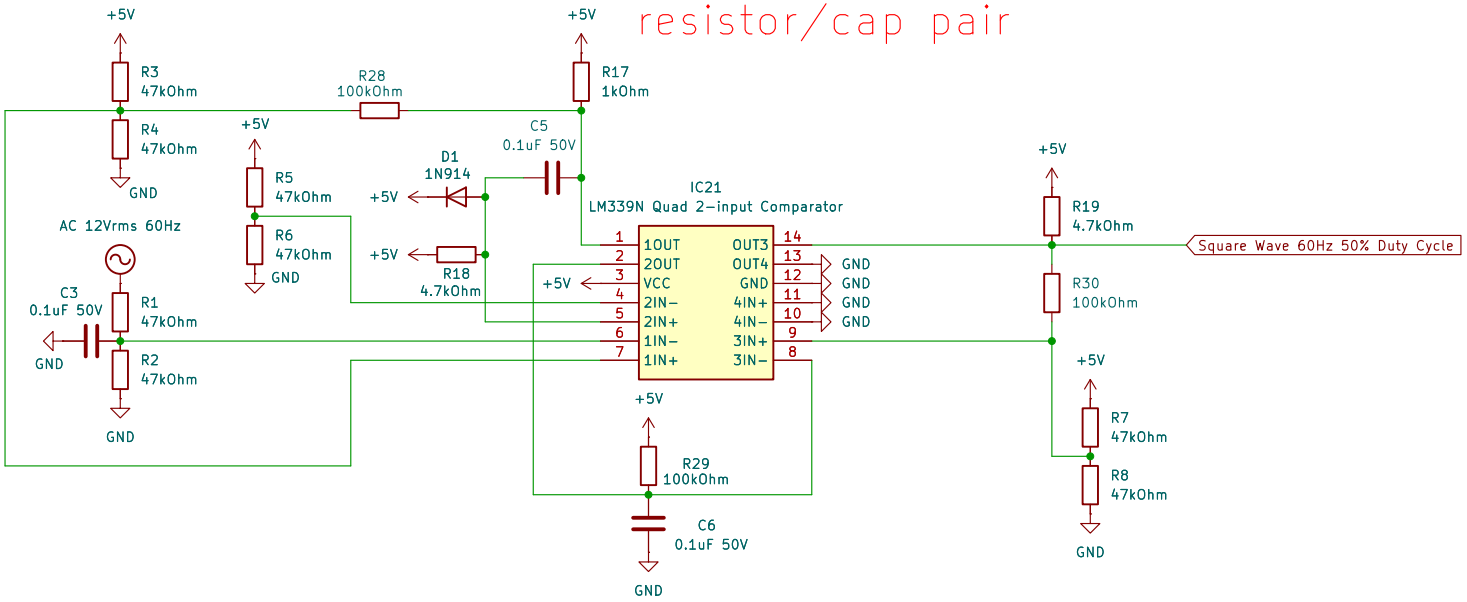
Sheet: /Power/
File: Power.kicad_sch

Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 2/10

Square Wave Generator

Supposed to be 1koh??
or was this the time constant
resistor/cap pair

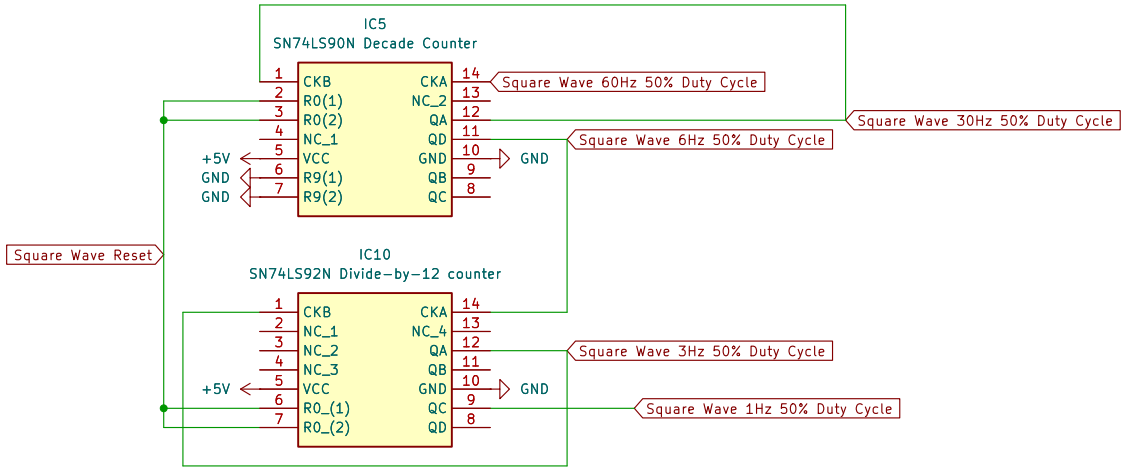


https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Square Wave Generator/
File: Square_Wave_Generator.kicad_sch

Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/10



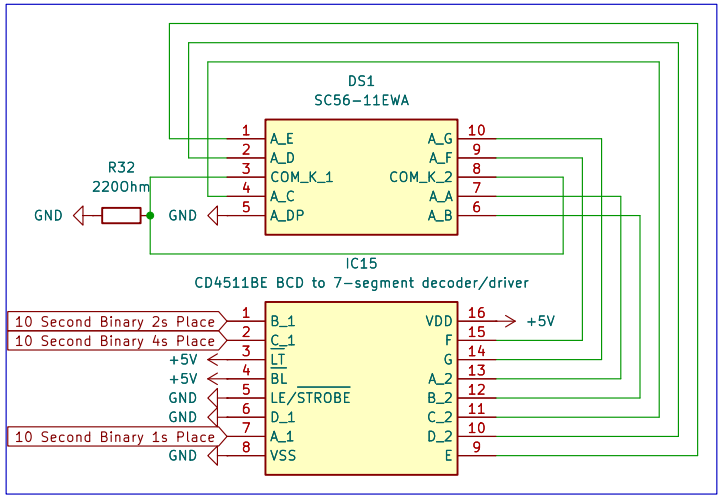
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /1Hz Square Wave Generator/
File: 1Hz_Square_Wave_Generator.kicad_sch

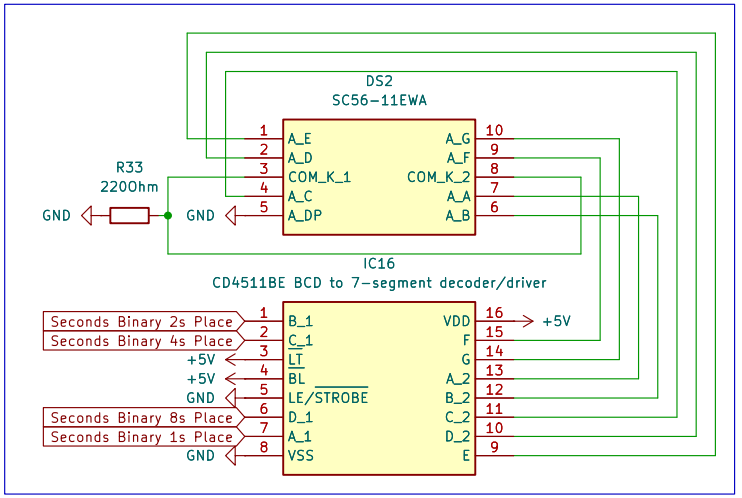
Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/10

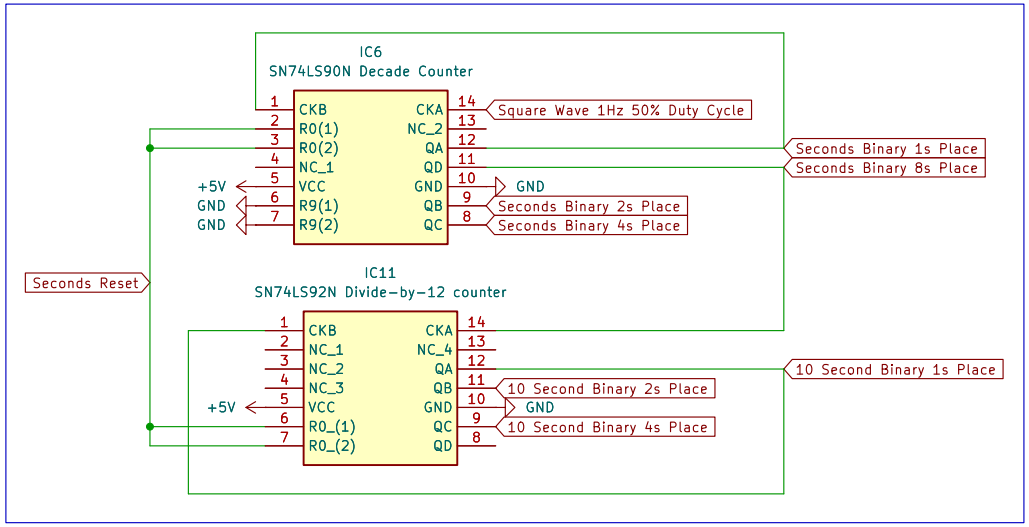
Seconds 10s Display



Seconds 1s Display



Seconds Counter



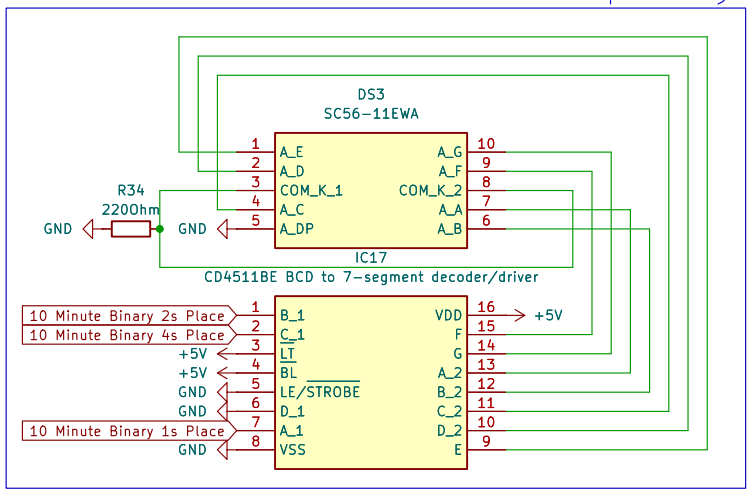
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
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Sheet: /Seconds Counter/
File: Seconds_Counter.kicad_sch

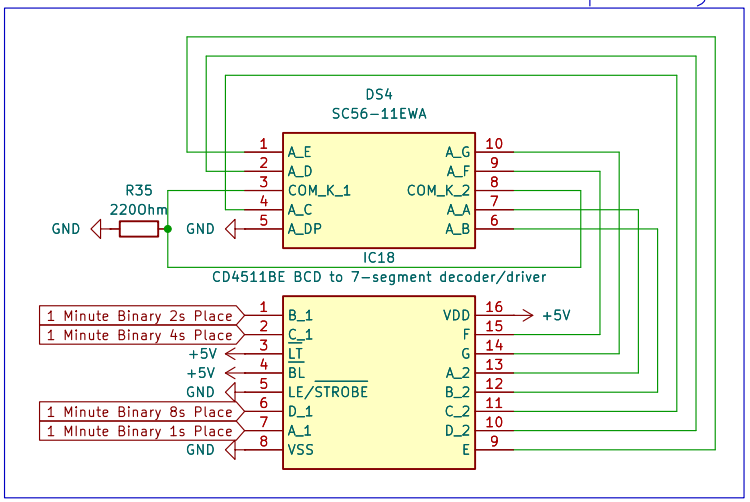
Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/10

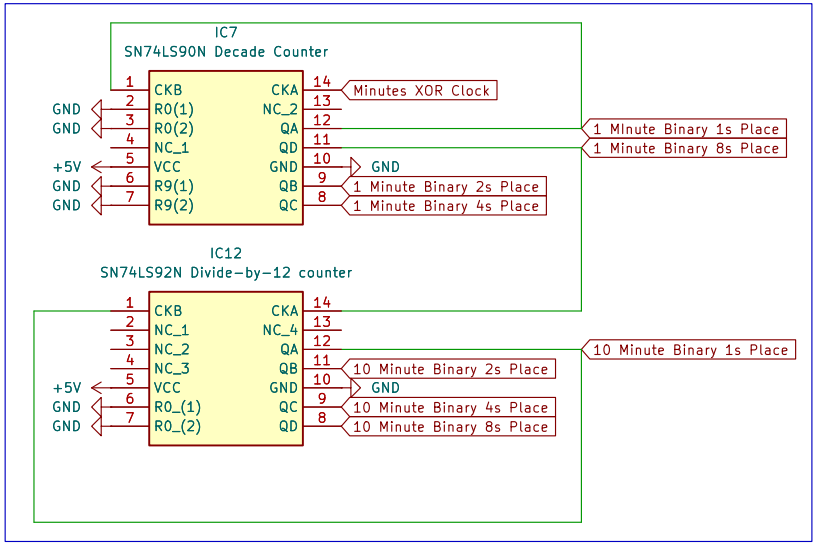
Minutes 10s Display



Minutes 1s Display



Minutes Counter



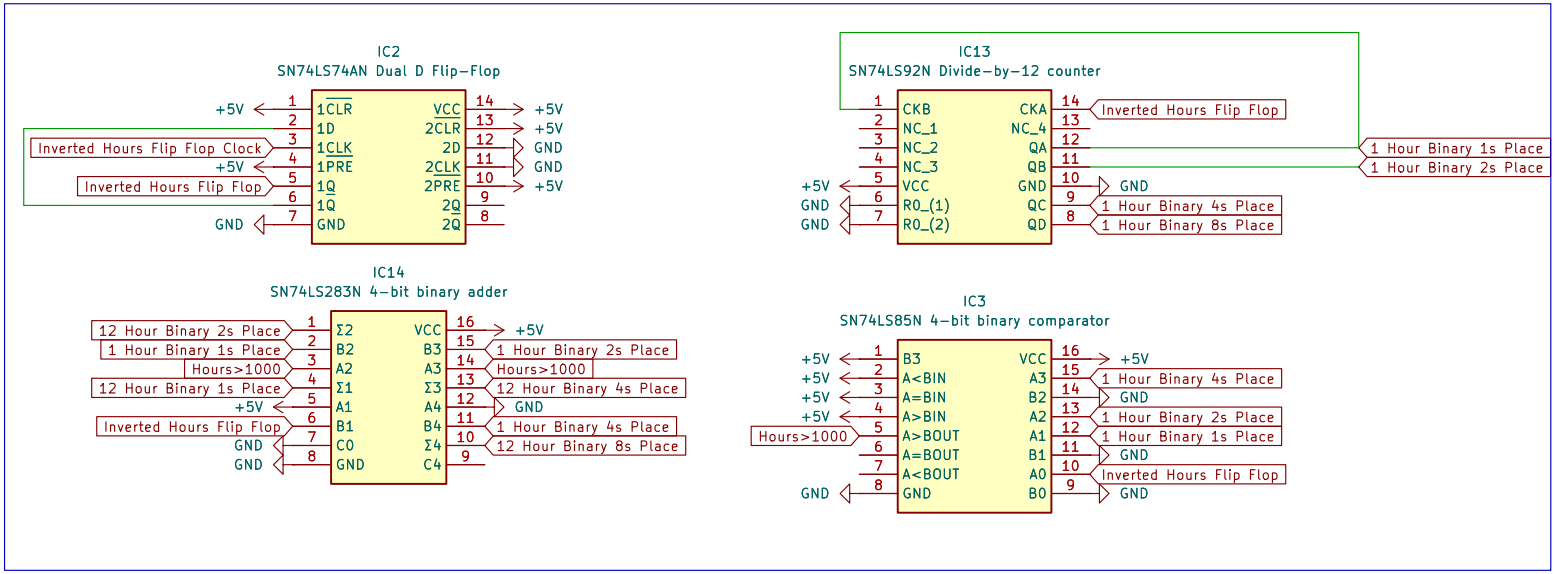
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Minutes Counter/
File: Minutes_Counter.kicad_sch

Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 6/10

Hours Counter



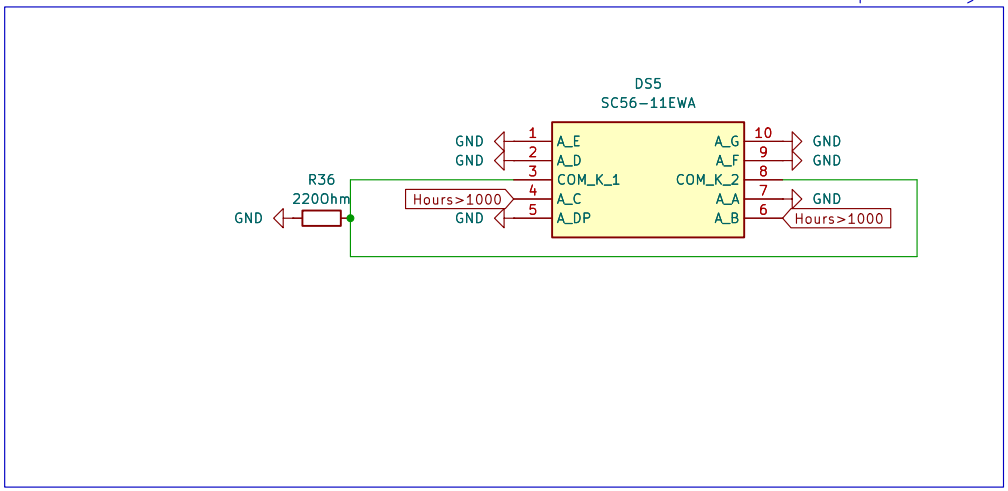
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Hours Counter/
File: Hours_Counter.kicad_sch

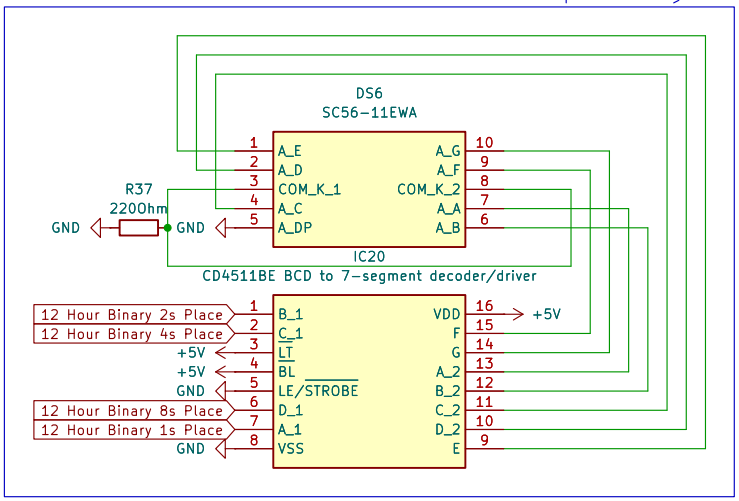
Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 7/10

Hours 10s Display



Hours 1s Display



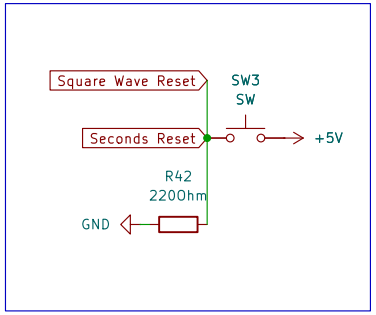
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Hours Display/
File: Hours_Display.kicad_sch

Title: 12 Hour Breadboard Clock

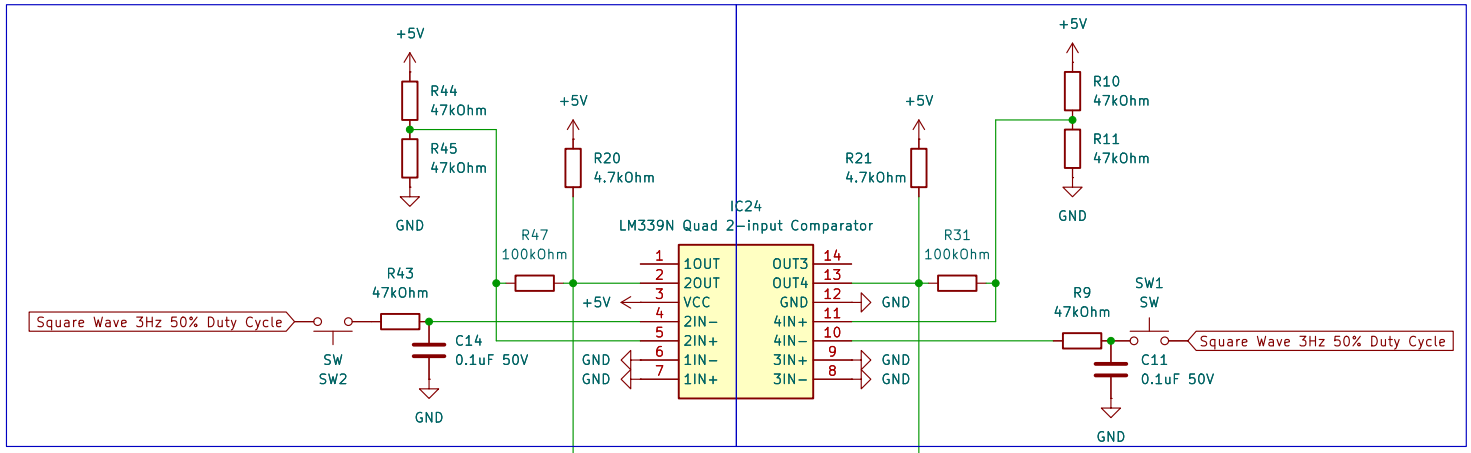
Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 8/10

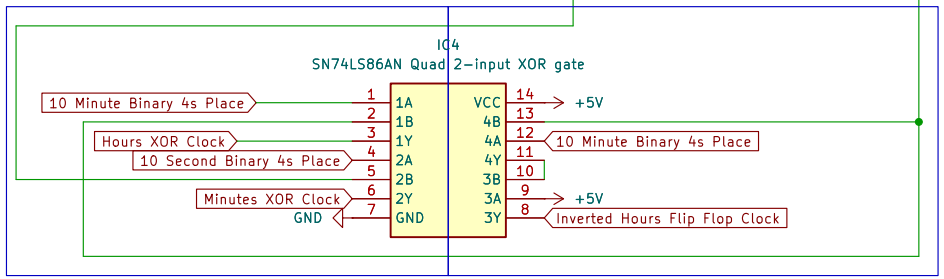
Seconds and Square Wave Reset



Minutes 3Hz Set Button

Hours 3Hz Set Button





Hour Clock Inverter

Hour/Minute XOR Logic

Still needed?????

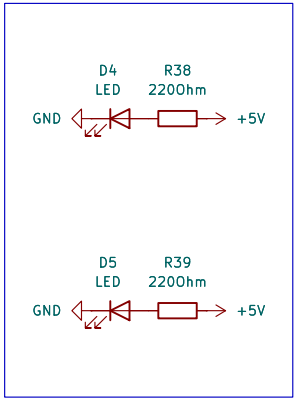
https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Time Set Buttons/
File: Time_Set_Buttons.kicad_sch

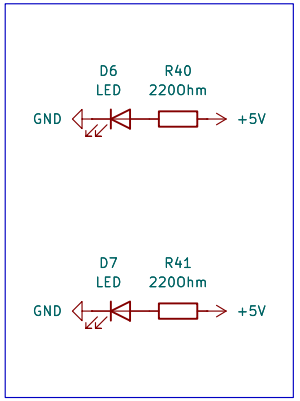
Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 9/10

Hours/ Minutes Colon



Minutes/ Seconds Colon



https://www.youtube.com/watch?v=3XDH-fZKnQk&t=629s
Design made from Ben Eater's YouTube series
KICAD Schematics by Jost Strnat

Sheet: /Colon LEDs/
File: Colon LEDs.kicad_sch

Title: 12 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 10/10